The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 September 2018.

INCH-POUND

MIL-PRF-19500/676G
7 June 2018
SUPERSEDING
MIL-PRF-19500/676F
9 May 2014

PERFORMANCE SPECIFICATION SHEET

* TRANSISTOR, FIELD EFFECT RADIATION HARDENED, N-CHANNEL, SILICON, TYPES 2N7465 AND 2N7466, JANTXVR AND JANSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode, MOSFET, radiation hardened, power transistor. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy maximum rating ($E_{AS}$) and maximum avalanche current ($I_{AS}$). Provisions for radiation hardness assurance (RHA) to one radiation level "R" is provided for JANS and JANTXV product assurance levels.

* 1.2 Package outlines. The device package outlines are as follows: TO-257AA in accordance with figure 1 and TO-276AA surface mount in accordance with figure 2 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| Type | $P_T$ (1) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C (free air) | $R_{\theta JC}$ (2) | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $I_{D1}$ (3) (4) $T_C$ = +25°C | $I_{D2}$ (3) (4) $T_C$ = +100°C | $I_S$ | $I_{DM}$ (5) | $T_J$ and $T_{STG}$ | $V_{ISO}$ 70,000 foot altitude |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | V dc | V dc | V dc | A dc | A dc | A dc | A(pk) | °C | V dc |
| 2N7465T3, U3 | 75 | 2 | 1.67 | 400 | 400 | ±20 | 5.0 | 3.2 | 5.0 | 20 | -55 to +150 | 400 |
| 2N7466T3, U3 | 75 | 2 | 1.67 | 500 | 500 | ±20 | 4.4 | 2.8 | 4.4 | 17.6 | | 500 |

(1) Derate linearly 0.6 W/°C for $T_C$ > +25°C.
(2) See figure 3, thermal impedance curves.
(3) The following formula derives the maximum theoretical $I_D$ limit. $I_D$ is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

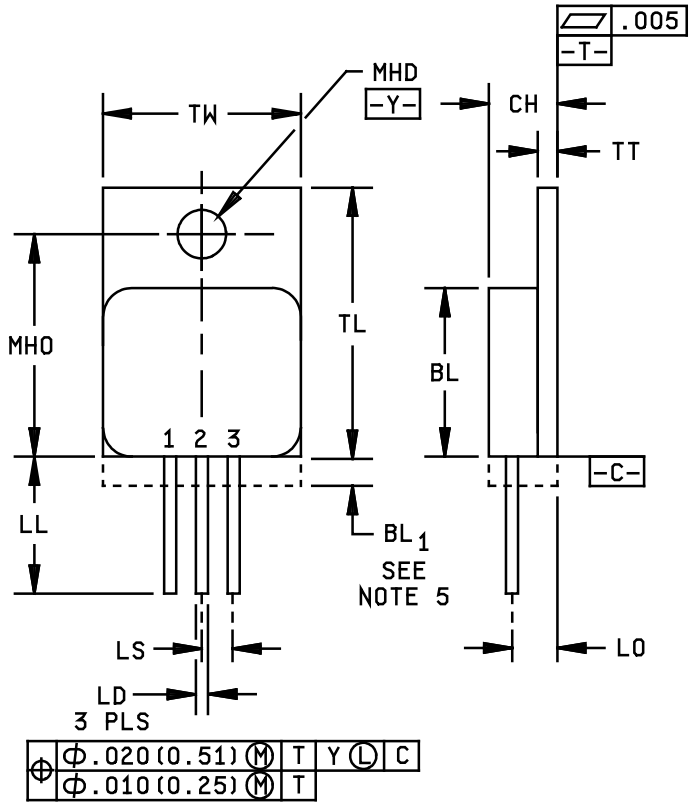
(4) See figure 3, maximum drain current graphs.
(5) $I_{DM}$ = 4 X $I_{D1}$ as calculated in note (3).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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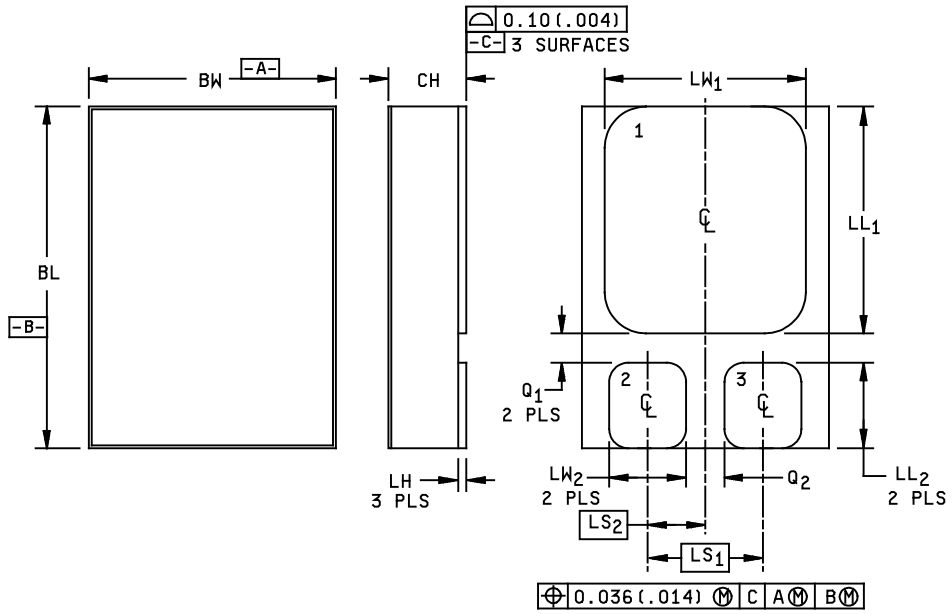

| Ltr | Inches | | Millimeters | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| BL | .410 | .430 | 10.41 | 10.92 |
| $BL_1$ | | .033 | | 0.84 |
| CH | .190 | .200 | 4.83 | 5.08 |
| LD | .025 | .035 | 0.64 | 0.89 |
| LL | .600 | .650 | 15.24 | 16.51 |
| LO | .120 BSC | | 3.05 BSC | |
| LS | .100 BSC | | 2.54 BSC | |
| MHD | .140 | .150 | 3.56 | 3.81 |
| MHO | .527 | .537 | 13.39 | 13.64 |
| TL | .645 | .665 | 16.38 | 16.89 |
| TT | .035 | .045 | 0.89 | 1.14 |
| TW | .410 | .420 | 10.41 | 10.67 |
| Term 1 | Drain | | | |
| Term 2 | Source | | | |
| Term 3 | Gate | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. All terminals are isolated from the case.
4. This area is for the lead feed-thru eyelets (configuration is optional, but will not extend beyond this zone).
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-257AA (2N7465T3 and 2N7466T3).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .395 | .405 | 10.04 | 10.28 |
| BW | .291 | .301 | 7.40 | 7.64 |
| CH | | .124 | | 3.15 |
| LH | .010 | .020 | 0.25 | 0.51 |
| LW1 | .281 | .291 | 7.14 | 7.39 |
| LW2 | .090 | .100 | 2.29 | 2.54 |
| LL1 | .220 | .230 | 5.59 | 5.84 |
| LL2 | .115 | .125 | 2.93 | 3.17 |
| LS1 | .150 BSC | | 3.81 BSC | |
| LS2 | .075 BSC | | 1.91 BSC | |
| Q1 | .030 | | 0.76 | |
| Q2 | .030 | | 0.76 | |
| TERM 1 | Drain | | | |
| TERM 2 | Gate | | | |
| TERM 3 | Source | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for TO-276AA (2N7465U3 and 2N7466U3).



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1.4 Primary electrical characteristics at $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 $I_D$ = 1.0 mA dc | $V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ $I_D$ = 1.0 mA dc | | Max $I_{DSS1}$ $V_{GS}$ = 0 $V_{DS}$ = 80 percent of rated $V_{DS}$ | Max $r_{DS(ON)}$ (1) $V_{GS}$ = 12 V dc | | $E_{AS}$ at $I_{D1}$ | $I_{AS}$ |
|---|---|---|---|---|---|---|---|---|
| | | | | | $T_J$ = +25°C at $I_{D2}$ | $T_J$ = +150°C at $I_{D2}$ | | |
| | V dc | V dc | | µA dc | ohm | ohm | mJ | A |
| | | Min | Max | | | | | |
| 2N7465T3, U3 | 400 | 2.5 | 4.5 | 50 | 1.39 | 3.0 | 150 | 5.0 |
| 2N7466T3, U3 | 500 | 2.5 | 4.5 | 50 | 1.77 | 3.9 | 150 | 4.4 |

(1) Pulsed (see 4.5.1).

* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JANTXV” and "JANS".

* 1.5.2 Radiation hardness assurance (RHA) designator. The RHA level that is applicable for this specification sheet is level "R".

* 1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

* 1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

* 1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "7465" and "7466".

* 1.5.3.3 Suffix letters. The suffix letters "T3" are used on devices that are packaged in the TO-257AA package of figure 1. The suffix letters "U3" are used on devices that are packaged in the TO-276AA package of figure 2.

* 1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil/

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

$I_{AS}$..... Rated avalanche current, non-repetitive.
nC ..... nano Coulomb.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (TO-257AA) and 2 (U3, surface mount, TO-276AA) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted to meet the requirements of this specification.



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* 3.5 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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4.2.1.1 Single event effects (SEE). SEE shall be performed at initial qualification and after process or design changes which may affect radiation hardness (see table III and table IV). Upon qualification, manufacturers shall provide the verification test conditions from section 5 of method 1080 of MIL-STD-750 that were used to qualify the device for inclusion into section 6 of the performance specification sheet. End-point measurements shall be in accordance with table II. SEE characterization data shall be made available upon request of the qualifying or acquiring activity.

* 4.3 Screening (JANS and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) (1) (2) | Measurement | |
|---|---|---|
| | JANS level | JANTXV level |
| (3) | Gate stress test (see 4.3.1) | Gate stress test (see 4.3.1) |
| (3) | Method 3470 of MIL-STD-750, $E_{AS}$ (see 4.3.2) | Method 3470 of MIL-STD-750, $E_{AS}$ (see 4.3.2) |
| (3) 3c | Method 3161 of MIL-STD-750, thermal impedance (see 4.3.3) | Method 3161 of MIL-STD-750, thermal impedance (see 4.3.3) |
| 9 | Subgroup 2 of table I herein | Not applicable |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| 11 | Subgroup 2 of table I herein,<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 μA dc or ±100 percent of initial value, whichever is greater. | Subgroup 2 of table I herein |
| 12 | Method 1042 of MIL-STD-750, test condition A | Method 1042 of MIL-STD-750, test condition A |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 μA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$= ±20 percent of initial value. | Subgroup 2 of table I herein;<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±10 μA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$= ±20 percent of initial value. |
| 17 | For TO–257AA and TO-276AA packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein | For TO–257AA and TO-276AA packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ are measured.
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTXV does not need to be repeated in screening requirements.



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4.3.1 Gate stress test. Apply $V_{GS}$ = 30 V minimum for t = 250 μs minimum.

4.3.2 Single pulse avalanche energy ($E_{AS}$).

a. Peak current ($I_{AS}$) ..... $I_{AS(max)}$.

b. Peak gate voltage ($V_{GS}$) ..... 12 V.

c. Gate to source resistor ($R_{GS}$) ..... $25\Omega \leq R_{GS} \leq 200\Omega$.

d. Initial case temperature ($T_C$) ..... +25°C +10°C, -5°C.

e. Inductance (L)..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Supply voltage ($V_{DD}$) ..... 50 V.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, (and $V_H$ where appropriate). (See figure 4 herein.) Measurement delay time ($t_{MD}$) = 70 μs max. See table III, group E, subgroup 4 herein.

4.3.4 Dielectric withstanding voltage.

a. Magnitude of test voltage..... 800 V dc (TO-257AA); 900 V dc (TO-276AA).

b. Duration of application of test voltage..... 15 seconds (min).

c. Points of application of test voltage ..... All leads to case (bunch connection).

d. Method of connection ..... Mechanical.

e. Kilovolt-ampere rating of high voltage source ..... 1,200 V/1.0 mA (min).

f. Maximum leakage current..... 1.0 mA.

g. Voltage ramp up time..... 500 V/second

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and herein.



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* 4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1051 | Test condition G, 100 cycles. |
| B3 | 2075 | See 3.4.2 herein. |
| B3 | 2077 | Scanning electron microscope (SEM) qualification may be performed anytime prior to lot formation. |
| B4 | 1042 | Condition D. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 60 seconds minimum. |
| B5 | 1042 | Test condition B, $V_{GS}$ = rated $T_A$ = +175°C, t = 24 hours. |
| B5 | 1042 | Condition A, $V_{DS}$ = rated; $T_A$ = +175°C; t = 120 hours. |

* 4.4.2.2 Quality levels JANTXV, table E-VIB of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | Test condition G, 25 cycles. (45 total, including 20 cycles performed in screening) |
| B3 | 1042 | Test condition D. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 60 seconds minimum. |
| B4 | 2075 | See 3.4.2 herein. |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 1056 | Test condition B. |
| C2 | 2036 | Test condition A, weight = 10 lbs (4.54 Kg), t = 10 s (applicable to TO-257AA only). |
| C5 | 3161 | See 4.3.3, $R_{\theta JC}$ = 1.67 °C/W. |
| C6 | 1042 | Test condition D, 6,000 cycles. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 60 seconds minimum. |

4.4.4 Group D Inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

* 4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3161 | See 4.3.3 | $Z_{\theta JC}$ | | | °C/W |
| Breakdown voltage, drain to source | 3407 | $V_{GS}$ = 0, $I_D$ = 1 mA dc, bias condition C | $V_{(BR)DSS}$ | | | |
| 2N7465T3, U3 | | | | 400 | | V dc |
| 2N7466T3, U3 | | | | 500 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)1}$ | 2.5 | 4.5 | V dc |
| Gate reverse current | 3411 | $V_{GS}$ = +20 V dc bias condition C, $V_{DS}$ = 0 | $I_{GSSF1}$ | | +100 | nA dc |
| Gate reverse current | 3411 | $V_{GS}$ = -20 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSSR1}$ | | -100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0 , bias condition C, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS1}$ | | 50 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D$ = $I_{D2}$ | $r_{DS(on)}1$ | | | |
| 2N7465T3, U3 | | | | | 1.39 | Ω |
| 2N7466T3, U3 | | | | | 1.77 | Ω |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D$ = $I_{D1}$ | $r_{DS(on)}2$ | | | |
| 2N7465T3, U3 | | | | | 1.52 | Ω |
| 2N7466T3, U3 | | | | | 1.90 | Ω |
| * Forward voltage | 4011 | Condition A, $I_D$ = $I_{D1}$, $V_{GS}$ = 0 | $V_{SD}$ | | 1.2 | V |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_C = T_J = +125°C$ | | | | |
| Gate reverse current | 3411 | $V_{GS}$ = +20 V dc and -20 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0, bias condition C, $V_{DS}$ = 100 percent of rated $V_{DS}$ | $I_{DSS2}$ | | 1.0 | mA dc |
| Drain current | 3413 | $V_{GS}$ = 0, bias condition C, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS3}$ | | 0.25 | mA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, bias condition A, pulsed (see 4.5.1), $I_D = I_{D2}$ | $r_{DS(on)3}$ | | | |
| 2N7465T3, U3 | | | | | 2.64 | Ω |
| 2N7466T3, U3 | | | | | 3.76 | Ω |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)2}$ | 1.5 | | V dc |
| Low-temperature operation: | | $T_C = T_J = -55°C$ | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)3}$ | | 5.5 | V dc |
| Subgroup 4 | | | | | | |
| Forward transconductance | 3475 | $I_D$ = rated $I_{D2}$, $V_{DD}$ = 15 V (see 4.5.1) | $g_{FS}$ | | | |
| 2N7465T3, U3 | | | | 0.5 | | mhos |
| 2N7466T3, U3 | | | | 0.4 | | mhos |
| Switching time test | 3472 | $I_D = I_{D1}$, $V_{GS}$= 12 V dc, $R_G$ = 7.5 Ω, $V_{DD}$ = 50 percent of rated $V_{DS}$ | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | | |
| 2N7465T3, U3 | | | | | 25 | ns |
| 2N7466T3, U3 | | | | | 25 | ns |
| Rise time | | | $t_r$ | | | |
| 2N7465T3, U3 | | | | | 75 | ns |
| 2N7466T3, U3 | | | | | 65 | ns |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 - Continued | | | | | | |
| Turn-off delay time | | | $t_{d(off)}$ | | | |
| 2N7465T3, U3 | | | | | 58 | ns |
| 2N7466T3, U3 | | | | | 60 | ns |
| Fall time | | | $t_f$ | | | |
| 2N7465T3, U3 | | | | | 58 | ns |
| 2N7466T3, U3 | | | | | 63 | ns |
| Subgroup 5 | | | | | | |
| Safe operating area test (high voltage) | 3474 | See figures 5 and 6; $t_p$ = 10 ms, $V_{DS}$ = 200 V | | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B | $Q_{G(on)}$ | | | |
| 2N7465T3, U3 | | | | | 31 | nC |
| 2N7466T3, U3 | | | | | 30 | nC |
| Gate to source charge | | | $Q_{GS}$ | | | nC |
| 2N7465T3, U3 | | | | | 8.5 | nC |
| 2N7466T3, U3 | | | | | 8 | nC |
| Gate to drain charge | | | $Q_{GD}$ | | | |
| 2N7465T3, U3 | | | | | 20 | nC |
| 2N7466T3, U3 | | | | | 18 | nC |
| Reverse recovery time | 3473 | Condition A, di/dt ≤ 100A/µs, $V_{DD}$ ≤ 50 V, $I_D = I_{D1}$ | $t_{rr}$ | | | |
| 2N7465T3, U3 | | | | | 350 | ns |
| 2N7466T3, U3 | | | | | 400 | ns |

1/ For sampling plan, see MIL-PRF-19500.

* 2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



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* TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Preirradiation limits | | Postirradiation limits | | Unit |
|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | R | | R | | |
| | | | | Min | Max | Min | Max | |
| Subgroup 1 | | | | | | | | |
| Not applicable | | | | | | | | |
| Subgroup 2 | | $T_C$ = +25°C | | | | | | |
| Steady-state total dose irradiation ($V_{GS}$ bias) 4/ | 1019 | $V_{GS}$ = 12V dc<br>$V_{DS}$ = 0 | | | | | | |
| Steady-state total dose irradiation ($V_{DS}$ bias) 4/ | 1019 | $V_{GS}$ = 0 V<br>$V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | | | | | | |
| End-point electricals: | | | | | | | | |
| Breakdown voltage, drain to source | 3407 | $V_{GS}$ = 0 V<br>$I_D$ = 1 mA<br>bias condition C | $V_{(BR)DSS}$ | | | | | |
| 2N7465T3, U3<br>2N7466T3, U3 | | | | 400<br>500 | | 400<br>500 | | V dc<br>V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$ | $V_{GS(th)1}$ | | | | | |
| 2N7465T3, U3<br>2N7466T3, U3 | | | | 2.5<br>2.5 | 4.5<br>4.5 | 2.0<br>2.0 | 4.5<br>4.5 | V dc<br>V dc |
| Gate reverse current | 3411 | $V_{GS}$ = 20 V dc<br>$V_{DS}$ = 0<br>bias condition C | $I_{GSSF1}$ | | 100 | | 100 | nA dc |
| Gate reverse current | 3411 | $V_{GS}$ = -20 V dc<br>$V_{DS}$ = 0<br>bias condition C | $I_{GSSR1}$ | | -100 | | -100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0 V<br>bias condition C<br>$V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | $I_{DSS1}$ | | 50 | | 50 | µA dc |

See footnotes at end of table.



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* TABLE II. Group D inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Preirradiation limits | | Postirradiation limits | | Unit |
|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | R | | R | | |
| | | | | Min | Max | Min | Max | |
| Static drain to source on-state voltage | 3405 | $V_{GS}$ = 12 V condition A pulsed (see 4.5.1) $I_D = I_{D2}$ | $V_{DSon1}$ | | | | | |
| 2N7465T3, U3 | | | | | 4.726 | | 4.726 | V dc |
| 2N7466T3, U3 | | | | | 4.956 | | 4.956 | V dc |
| * Forward voltage source to drain diode | 4011 | Condition A, $V_{GS}$ = 0 $I_D = I_{D1}$ | $V_{SD}$ | | 1.2 | | 1.2 | V dc |

1/ For sampling plan, see MIL-PRF-19500.
2/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification sheet utilizing the same die design.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ Separate samples shall be pulled for each bias.



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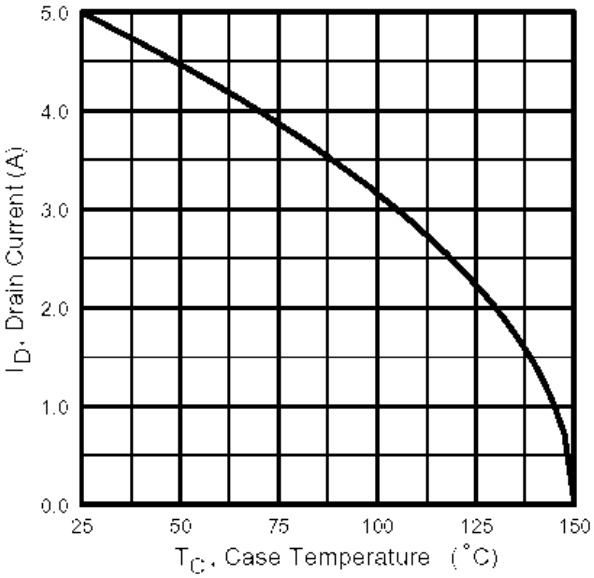
TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices c = 0 |
| Thermal shock (temperature cycling) | 1051 | Test condition G, 500 cycles | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 2 1/ | | | 45 devices c = 0 |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 4 | | | Sample size N/A |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 5 | | | 3 devices c = 0 |
| Barometric pressure (reduced) (all devices) | 1001 | $V_{DS}$ = rated $V_{(BR)DSS}$, $I_{(ISO)}$ < 0.25 mA | |
| Subgroup 10 | | | 22 devices c = 0 |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | Test conditions shall be derived by the manufacturer. | |
| Subgroup 11 | | | 3 devices |
| SEE 2/ 3/ | 1080 | See MIL-STD-750 method 1080. | |

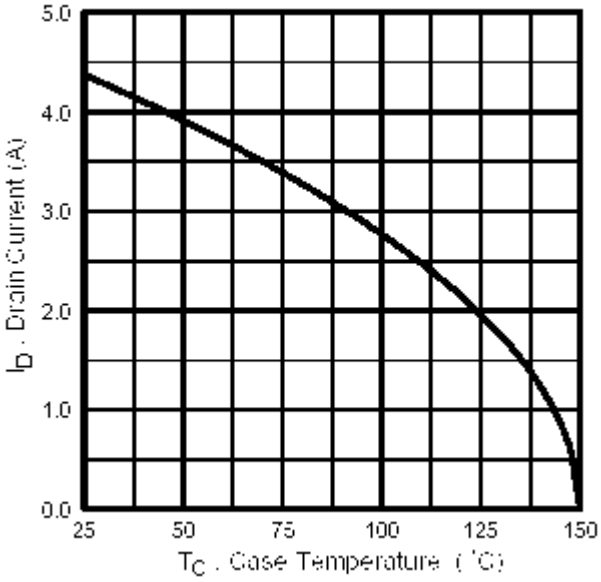
1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level LET is sufficient to qualify all lower level LETs.



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2N7465T3 and 2N7465U3

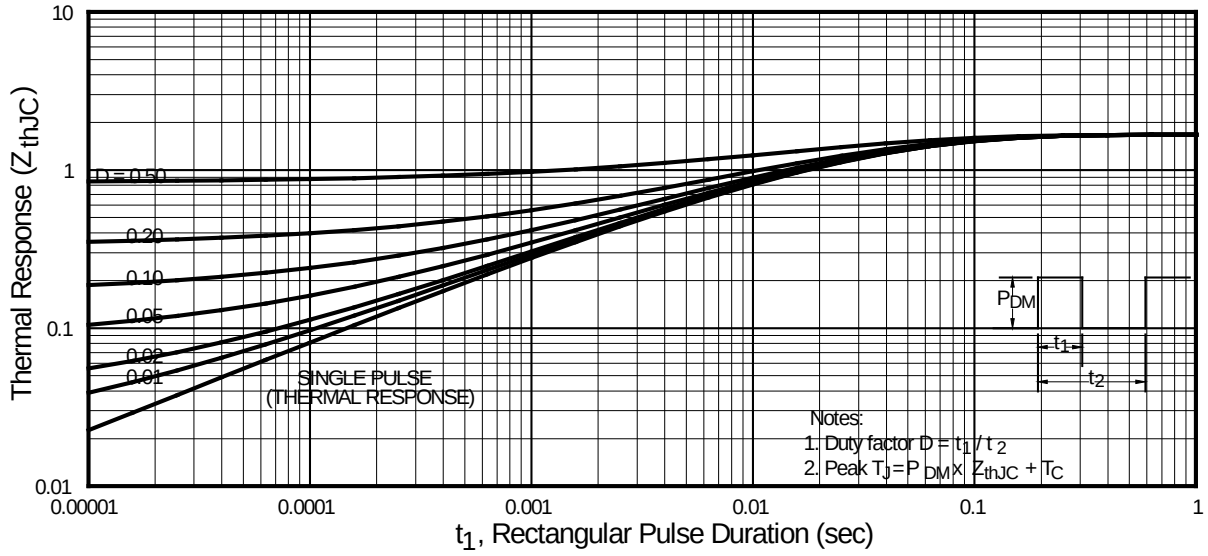



2N7466T3 and 2N7466U3

FIGURE 3. Maximum drain current and case temperature graphs.



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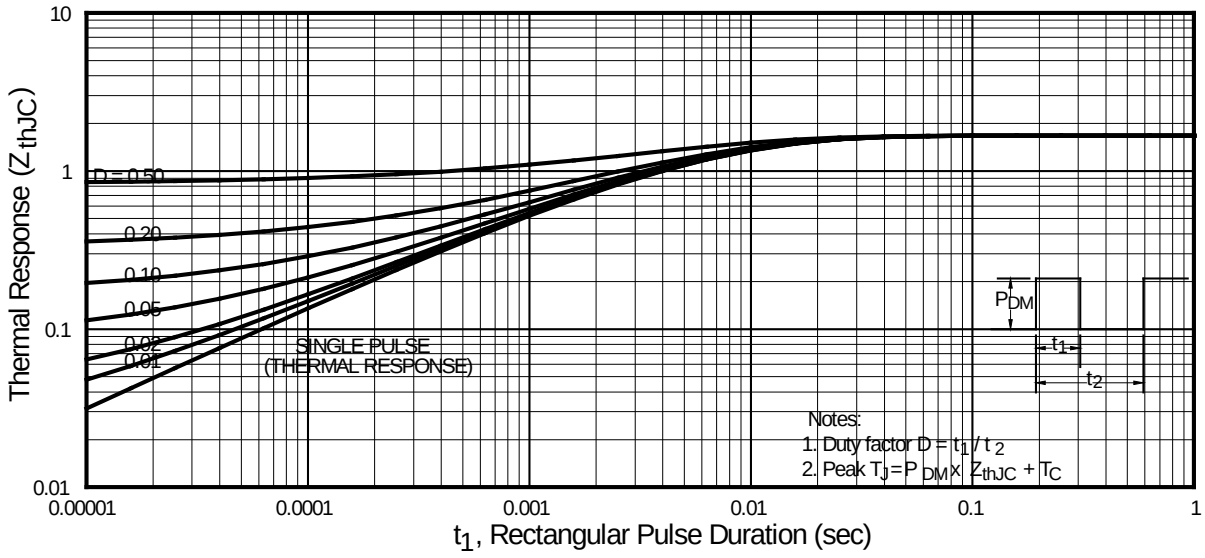



2N7465T3 and 2N7466T3

* FIGURE 4. Thermal impedance curves.



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2N7465U3 and 2N7466U3

* FIGURE 4. Thermal impedance curves – Continued.



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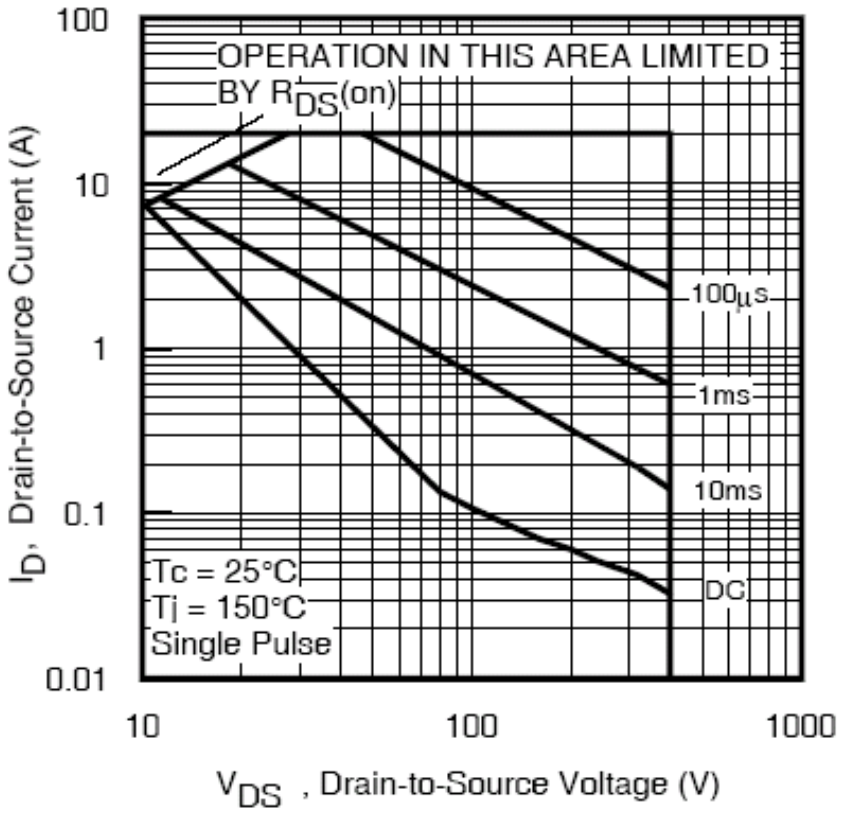



FIGURE 5. Safe operating area graph (2N7465T3 and 2N7465U3).



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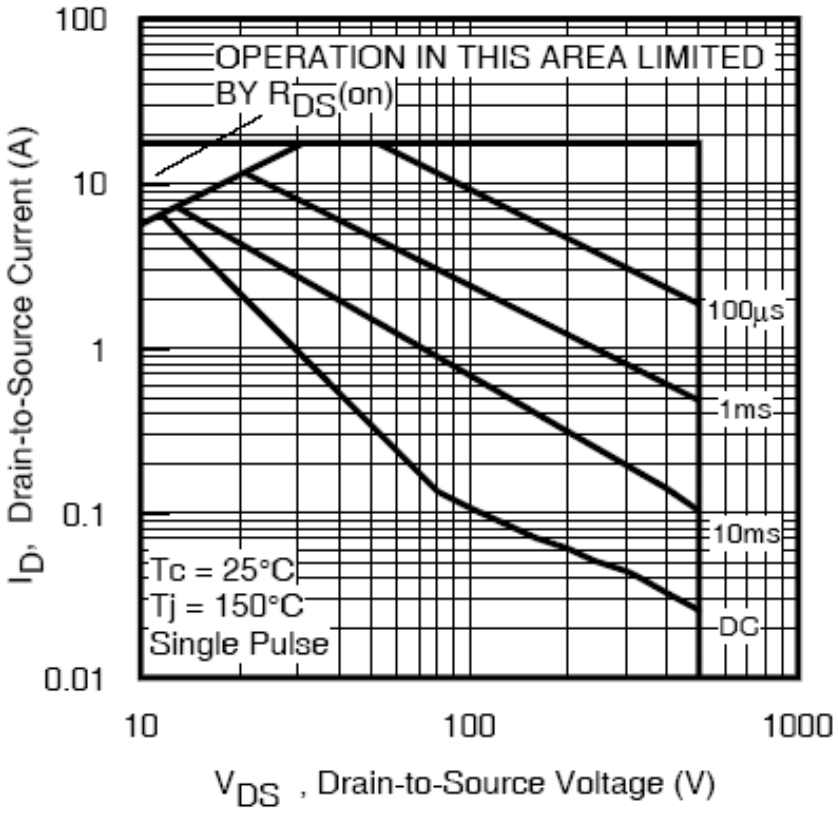



FIGURE 6. Safe operating area graph (2N7466T3 and 2N7466U3).



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

* d. The complete PIN, see 1.5 and 6.5.

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract or order.

f. If SEE testing data is desired, it should be specified in the contract or order.

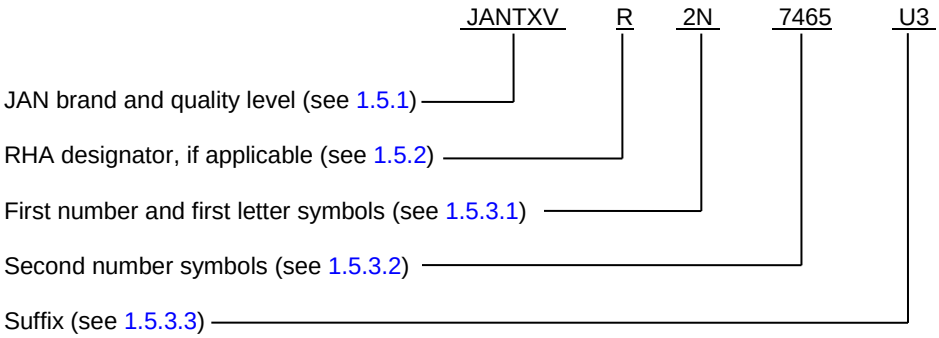
g. If specific SEE characterization conditions are desired (see section 6.7 and table IV), manufacturer's CAGE code should be specified in the contract or order.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.



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* 6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.




* 6.5 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for devices of the "TXV" quality level | PINs for devices of the "S" quality level |
|---|---|
| JANTXV#2N7465T3 | JANS#2N7465T3 |
| JANTXV#2N7465T3 | JANS#2N7465T3 |
| JANTXV#2N7466T3 | JANS#2N7466T3 |
| JANTXV#2N7466U3 | JANS#2N7466U3 |

* (1) The number sign (#) represent one RHA designator available (R). The PIN is also available without a RHA designator.

6.6 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

| Preferred types (military PIN) | Commercial PIN | |
|---|---|---|
| | TO-257AA | SMD.5 |
| 2N7465T3, U3<br>2N7466T3, U3 | IRHY7330CMSE<br>IRHY7430CMSE | IRHNJ7330SE<br>IRHNJ7430SE |

6.7 Application data.

6.7.1 Manufacturer specific irradiation data. Each manufacturer qualified to this slash sheet has characterized its devices to the requirements of MIL-STD-750 method 1080 and as specified herein. Since each manufacturer's characterization conditions can be different and can vary by the version of method 1080 qualified to, the MIL-STD-750 method 1080 revision version date and conditions used by each manufacturer for characterization have been listed here (see table IV) for information only. SEE conditions and figures listed in section 6 are current as of the date of this specification sheet, please contact the manufacturer for the most recent conditions.



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TABLE IV. Manufacturers characterization conditions.

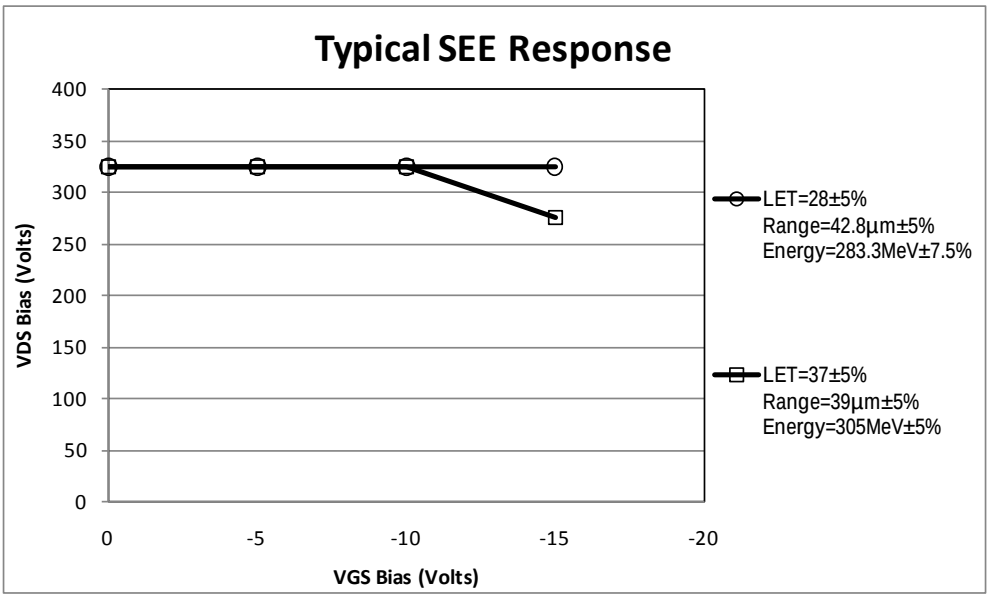
| Manufactures CAGE | Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|---|
| | | Method | Conditions | |
| 69210 (Applicable to devices with a date code of February 1998 and older) | SEE 1/ | 1080 | See MIL-STD-750E method 1080.0 dated 20 November 2006.<br>See figure 7. | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | 3 devices |
| | SEE irradiation: | | Fluence = 3E5 ±20 percent ions/cm$^2$<br>Flux = 2E3 to 2E4 ions/cm$^2$/sec, temperature = 25º ±5 °C | |
| | 2N7465T3 & 2N7465U3 | | Surface LET = 28 MeV-cm$^2$/mg ±5.0 %, range = 42.8 µm ±7.5%, energy = 283.3 MeV ±7.5%<br>In-situ bias conditions: $V_{DS}$ = 325 V and $V_{GS}$ = -15 V<br>(typical 4.53 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | 2N7466T3 & 2N7466U3 | | In-situ bias conditions: $V_{DS}$ = 375 V and $V_{GS}$ = -20 V<br>(nominal 4.53 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | 2N7465T3 & 2N7465U3 | | Surface LET = 37 MeV-cm$^2$/mg ±5.0 %, range = 39 µm ±7.5%, energy = 305 MeV ±7.5%<br>In-situ bias conditions: $V_{DS}$ = 325 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 275 V and $V_{GS}$ = -15 V<br>(nominal 3.77 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | 2N7466T3 & 2N7466U3 | | In-situ bias conditions: $V_{DS}$ = 350 V and $V_{GS}$ = -10 V<br>$V_{DS}$ = 325 V and $V_{GS}$ = -15 V<br>$V_{DS}$ = 300 V and $V_{GS}$ = -20 V<br>(typical 3.77 MeV/nucleon at Brookhaven National Lab Accelerator) | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | |
| Upon qualification, all manufacturers should provide the verification test conditions to be added to this table. | | | | |

1/ $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ was examined before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.



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2N7465T3, 2N7465U3




2N7466T3, 2N7466U3

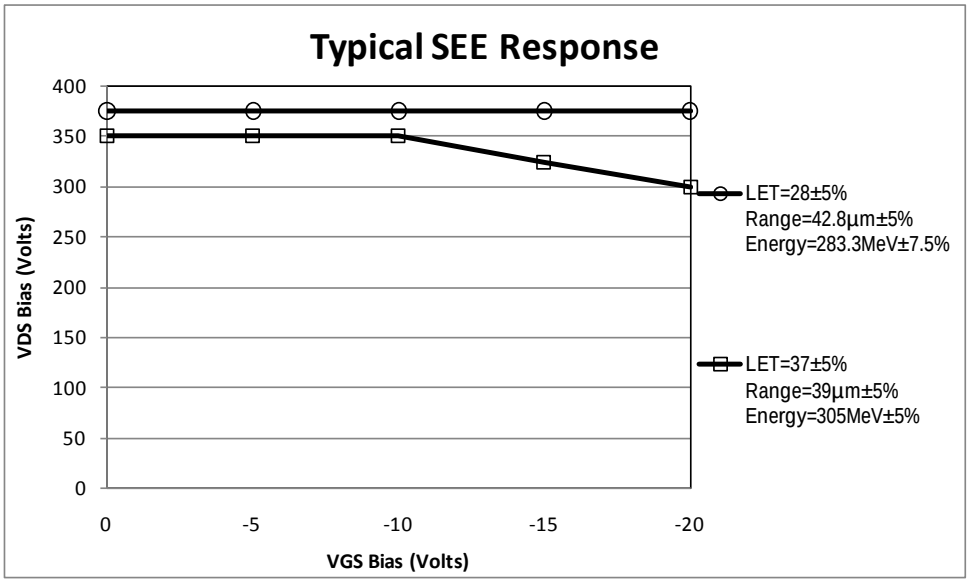



FIGURE 7. SEE safe operation area graph.



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* 6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.9 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - EC
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Review activity:
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